K-index Calculation Algorithm

Purpose and definition

This document describes the algorithm developed by Dr. Ingemar Haggstrom to compute local K-indices for the Kiruna magnetometer (IAGA code: KIR).

The K-index quantifies disturbances in the horizontal magnetic field components, X and Y, on a scale from 0 to 9. For each three-hour interval, calculate the range of the deflection from the presumed solar-quiet (SQ) value. The range is the maximum deflection minus the minimum deflection.

The overall sequence from monthly SQ refinement to three-hour K classification is summarized in Figure 1.

Kiruna K-index thresholds

K-index	Deflection range (nT)
0	0 to less than 15
1	15 to less than 30
2	30 to less than 60
3	60 to less than 120
4	120 to less than 210
5	210 to less than 360
6	360 to less than 600
7	600 to less than 990
8	990 to 1500, inclusive
9	greater than 1500

The source threshold table does not state how to handle shared boundaries. The intervals above make each shared boundary belong to the higher index, while honoring the source's explicit statement that K9 is greater than 1500 nT. Do not round the calculated range before classification.

Solar-quiet curve

The SQ curve represents the regular daily variation of the geomagnetic field during low solar activity. It consists of 24 values per component, one for each hour of the day. The curve is updated monthly.

The magnetometer records data every second. Interpolate the 24 hourly SQ values for X, Y, and Z to obtain an SQ value for every second.

Monthly SQ update

1. Read the previous month's 24 hourly SQ mean values for X, Y, and Z.
2. Interpolate the hourly values to one SQ value per second.
3. Select quiet intervals by comparing observations with the interpolated SQ values at the corresponding times. A time qualifies for the first pass when the difference is within 100 nT (the K3 threshold). Form a solar-quiet interval file from the qualifying times.
4. For each hour and component, take the median (or mean)[1] of the qualifying data to produce a new 24-value SQ file.
5. Interpolate this new hourly curve to one value per second.
6. Repeat the quiet-interval selection using a tighter difference of 10 nT (the K0 threshold).
7. For each hour and component, take the median (or mean)[1] of the qualifying data to produce the updated 24-value SQ file.
8. Use this file as the SQ curve for the next month.

[1]The median is recommended for high-latitude stations (such as Kiruna) because isolated substorms or auroral disturbances can occur even on quiet days and pull an arithmetic mean toward the disturbance.

The two passes use progressively tighter thresholds because the SQ curve is expected to vary by less than 10 nT from month to month. The first pass allows deviations up to 100 nT; the second refines the result using a 10 nT limit. Figure 2 illustrates how the filtered observations become the 24-row SQ file.

First-run bootstrap when no previous SQ file exists

The monthly update above assumes that a previous month's SQ file is available. On the first run, there is no such file, so first create a provisional seed curve from the observations. The original algorithm does not specify this initialization; the following is a recommended bootstrap procedure.

1. Collect an initial calibration window. A full month is a practical default because the SQ curve is updated monthly. Use the GFZ International Quiet and Disturbed Days list to identify candidate dates: Q1 is the quietest day of the month, through Q10, and D1-D5 are the most disturbed days. Prefer Q1-Q5 when they provide enough valid measurements; otherwise include additional Q days up to Q10. Exclude D1-D5 and data already identified as invalid.
2. Group the measurements by hour of day, using the same time basis and timestamp alignment that will be used for the operational SQ curve.
3. For each of the 24 hours and each component (X, Y, and Z), take the median (or mean)[1] of the valid measurements in that hour across the calibration window. These 72 hourly values form the provisional 24-hour SQ seed.
4. Interpolate the seed to one value per second and run the standard monthly SQ update above on the calibration data: first with the 100 nT criterion, then with the 10 nT criterion. Use the resulting curve as the initial operational SQ curve.
5. If an hour lacks enough valid observations to estimate a reliable hourly value, do not fill it with zero. Extend the calibration window or obtain a suitable SQ curve from another trusted source. Until all hourly values are available, mark the SQ curve and any K-indices that depend on it as unavailable or provisional.

The GFZ quiet/disturbed-day classification is based on planetary Kp criteria, not Kiruna-specific measurements. Treat it as a way to choose candidate dates, not as a substitute for the local 100 nT and 10 nT filters in the SQ update. The dataset is available from the GFZ Kp data page and is described as CC BY 4.0; cite Matzka et al. (2021), Geomagnetic Kp index, version 1.0, when using it.

The calibration-window length and minimum number of observations per hour are not specified by the source algorithm and should be documented by the implementation.

K-index calculation

For each day, calculate K-indices in eight consecutive three-hour intervals:

- 00:00-03:00
- 03:00-06:00
- 06:00-09:00
- 09:00-12:00
- 12:00-15:00
- 15:00-18:00
- 18:00-21:00
- 21:00-24:00

For each interval, subtract the interpolated SQ value from each observed horizontal component (X and Y). Find the maximum and minimum deflection in the interval and subtract the minimum from the maximum. Classify the resulting deflection range using the threshold table above.

The source describes the X and Y calculations but does not explicitly state how their two ranges are combined into a single reported K-index. That combination rule must be specified by the implementation.

[1] The median is recommended for high-latitude stations (such as Kiruna) because isolated substorms or auroral disturbances can occur even on quiet days and pull an arithmetic mean toward the disturbance.

Example solar-quiet file

The source document's example shows 24 hourly mean values. The columns are X, Y, and Z in pT.

Hour	X (pT)	Y (pT)	Z (pT)
00	10638894	-91957	51817732
01	10641294	-92162	51817152
02	10643882	-92377	51817457
03	10643891	-92356	51817470
04	10641757	-92354	51818139
05	10642735	-93584	51817495
06	10641265	-95282	51817208
07	10641714	-95960	51814396
08	10638799	-94869	51817234
09	10637983	-95847	51817535
10	10636815	-99162	51818190
11	10636732	-99002	51818487
12	10641875	-93074	51812544
13	10641863	-93029	51812523
14	10641310	-98623	51820069
15	10644902	-97646	51816755
16	10640593	-91926	51807756
17	10646601	-96004	51818344
18	10646277	-95996	51818349
19	10646277	-95632	51817617
20	10644188	-94023	51816800
21	10644110	-90536	51810423
22	10637333	-91737	51806640
23	10637400	-91740	51806645

Example solar-quiet interval file

Start time (YYYYMMDDHH)	End time (YYYYMMDDHH)
2016030223	2016030300
2016030317	2016030321
2016030418	2016030501
2016030502	2016030504
2016030600	2016030601
2016030603	2016030604
2016031302	2016031305
2016032217	2016032219
2016032316	2016032317
2016032517	2016032518
2016032521	2016032600
2016032616	2016032617
2016032816	2016032817
2016033122	2016033123

Figures

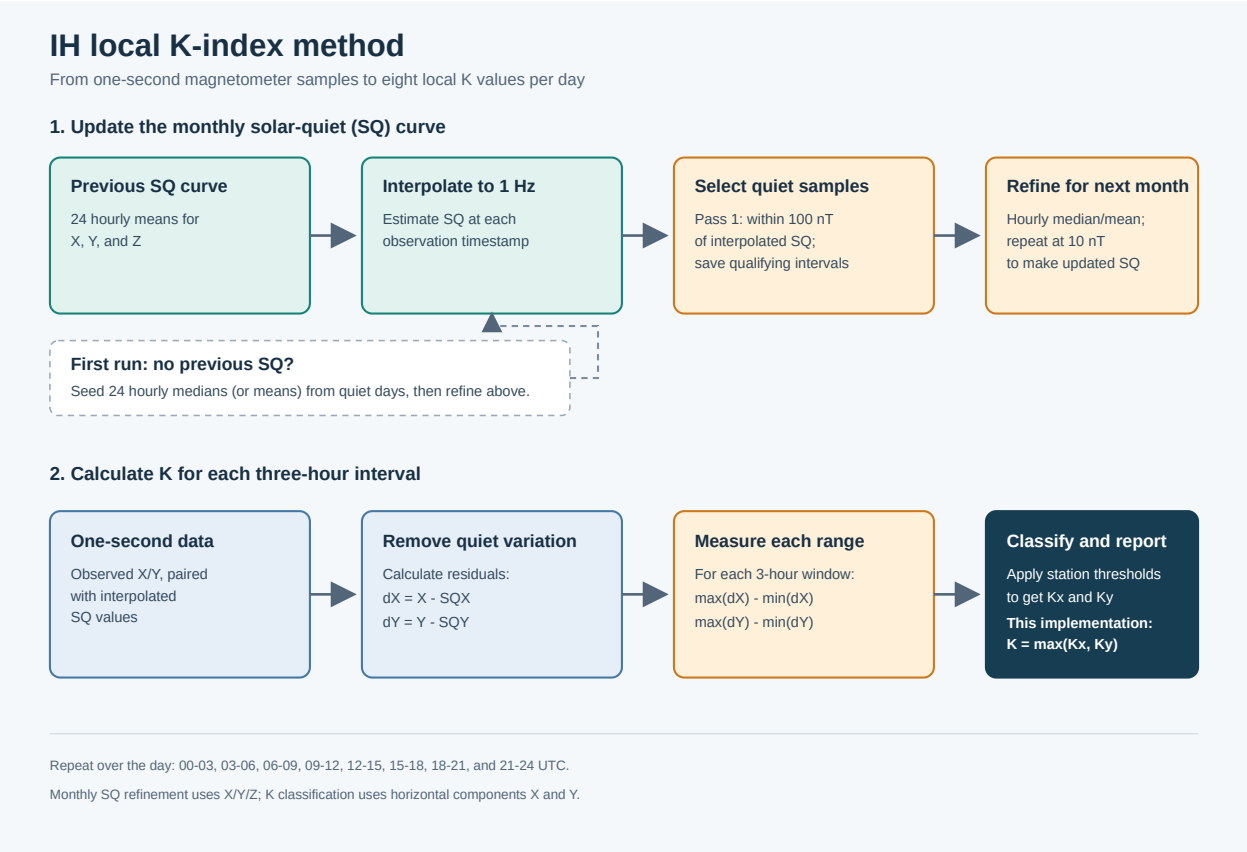


Figure 1. Overall IH method, from monthly SQ refinement through three-hour K classification.

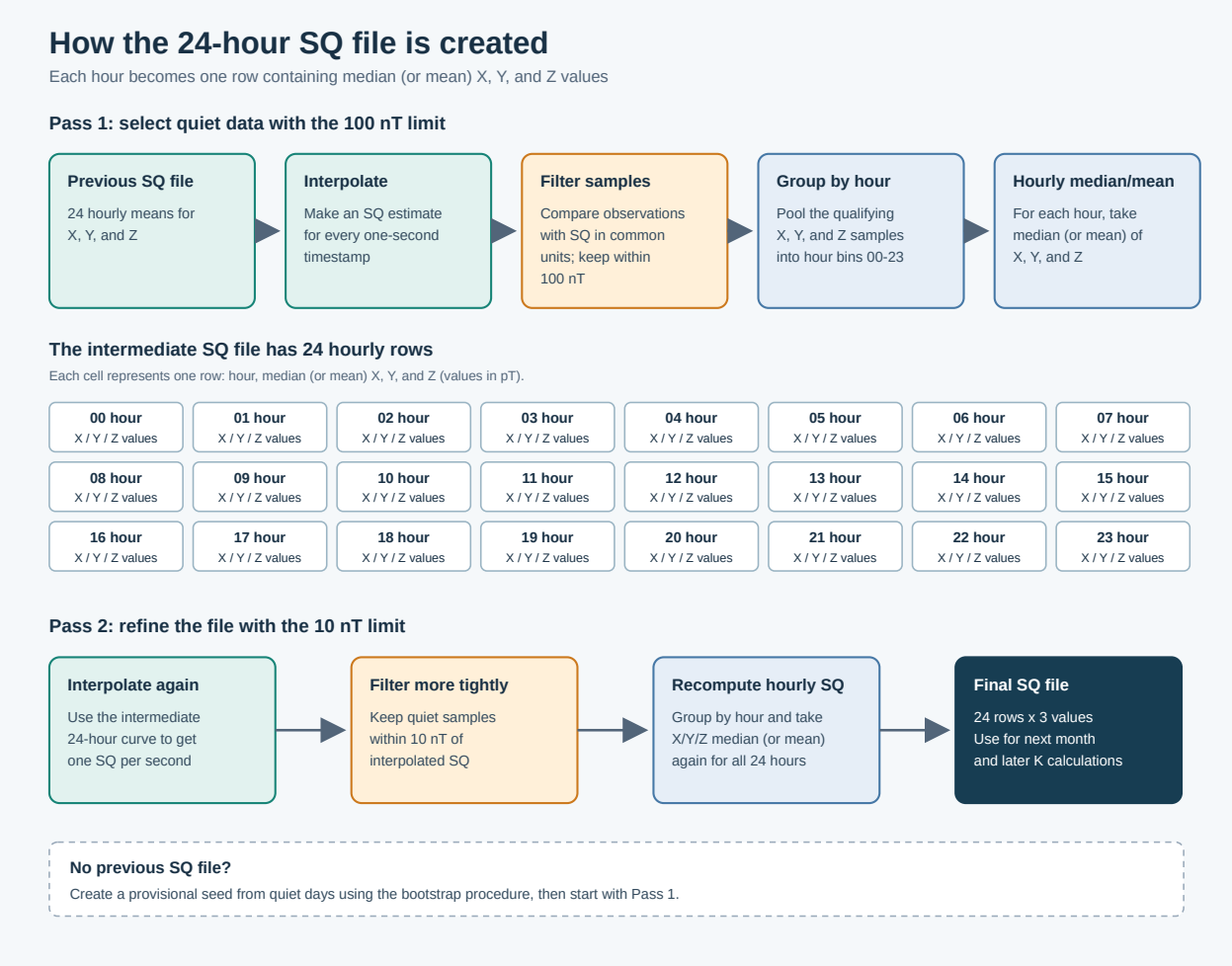


Figure 2. Creation of the 24-hour SQ file from filtered one-second X, Y, and Z observations.